

Mobility Management(Mandatory question in Exam from this topic Specifically Addressing in IP World)

Goals:

Continuous connectivity independent of location with high bandwidth, low latency, high reliability, guaranteed quality and at low cost

Challenge:

Transparent support of seamless mobility (seamless handoff) between different points of access to the Internet (from changing subnets)

Mobility Management in the reference Models:

1. What are the tasks of mobility management?

- Find and address a mobile away from home. Also called paging
- Support continuous communication after change of location
- Security – Avoid misuse of mechanism
- Privacy- hide location from others

2. Which protocol layer should handle the mobility problem? What are the implications of selecting one or another layer?

Each layer responds to different mobility scenarios.

Each layer has a specific task and is transparent to the layer above.

Network Layer (Layer 3)

Data Link Layer (Layer 2)

Transport Layer (Layer 4)

Handling mobility in wireless communication networks involves coordination between multiple protocol layers, but the specific layer primarily responsible for mobility management varies depending on the network architecture and mobility management approach.

3. What are the performance criteria for handovers?

In cellular telecommunications, handover, or handoff, is the process of transferring an ongoing call or data session from one channel connected to the core network to another channel. In satellite communications, it is the process of transferring satellite control responsibility from one earth station to another without loss or interruption of service.

Performance Criteria for Handovers:

- Handover Latency
- Handover success rate
- Packet loss and retransmission rate
- Signal strength and quality
- Deployment ease
- Security

4. Why can the cooperation between layers improve the handover performance?

Cooperation between layers can significantly improve handover performance by leveraging complementary functionalities, exchanging relevant information, and coordinating decision-making processes.

Link Layer Mobility Management (Layer 2 and PHY)

1) Which mobility problems can be handled on the link layer? Which not!

Mobility problems that can be handled on the link layer:

- Handovers within the same network
- Signal strength-based handover decision
- Address resolution and packet forwarding
- Localized Mobility Management

Mobility problems not handled on the link layer:

- Internetwork Handovers
- Global Mobility Management
- Session Continuity and quality of service

2) What happens if a mobile moves from one WLAN coverage area to another? What is the main source of handover delay?

Loss of signal, Scanning for new APS, Association with new APs, IP address renewal, and Handover completion.

The main source of handover delay in WLAN environments is typically the scanning process. Scanning Time, Channel switching, Authentication and association

3) How could this be accelerated?

Fast Roaming Protocols
Neighbour AP lists
Reduce Scan Times

Predictive handover
Channel pre-selection
Optimized Network Infrastructure
Dynamic Channel Assignment

Network Layer Mobility Management:

1) How can mobility be supported by the network layer?

It can be supported by the network layer in the following ways:

Mobile IP
Proxy Mobile
Hierarchical Mobile IPv6
Mobile Ad Hoc Networks
Dynamic Host Configuration
Local Management
Routing Optimization

2) When does the sole update of routing tables make sense and when not? Why does it not scale?

It can work in local updates or small networks. It does not make sense in global mobility. It does not scale because changing the IP tables for a single device in a large network doesn't make sense and adds a lot of signalling and overheads.

It makes sense in:

- Static environments
- Low mobility Environments
- Small scale networks

It doesn't make sense in:

- Highly dynamic environments
- Large Networks
- Complex topologies
- Resource constraints

3) What is the problem with changing the IP address for mobiles due to mobility?

Session Disruption
Reachability and Routing
Application Support
Security Considerations Scalability Management

4) What is the purpose of tunnelling? What is the advantage of this over specific routes?

Purpose of Tunnelling:

- IPv6 Transition

- VPNs
- Network Mobility
- Remote Access
- Routing Protocol Encapsulation

The advantage of tunnelling over specific routes lies in its flexibility, scalability, and compatibility

Mobile IP:

1) What is the overhead imposed on data traffic with Mobile IP?

Mobile IP introduces overhead to data traffic primarily due to encapsulation and additional signalling overhead.

- Encapsulation Overhead
- Tunnelling Overhead
- Header Expansion
- Signalling Overhead
- Handover Latency

2) What is the overhead for signalling with Mobile IP?

The overhead for signaling with Mobile IP primarily involves the exchange of control messages between the mobile node, home agent, and foreign agent to manage mobility and maintain session continuity.

Registration Signalling

Binding Update Signalling

Tunnel Establishment

Keep alive Signalling

Error Signalling

Hand over Signalling

The overhead for signaling with Mobile IP includes the exchange of registration messages, binding update messages, tunnel establishment messages, keepalive messages, error messages, and handover signaling messages. While signaling is essential for managing mobility and maintaining session continuity, it introduces additional protocol overhead, message exchanges, and network bandwidth consumption, which may impact overall network performance and scalability.

3) When is triangular routing especially harmful? Can we solve this and how?

Triangular routing occurs when the return path for packets is different from the forward path, leading to inefficient routing and potential performance degradation. Triangular routing is especially harmful in scenarios where latency-sensitive applications are used, such as real-time communication, video streaming, online gaming, or VoIP (Voice over Internet Protocol).

Here's why:

- Increased Latency
- Packet Loss
- Inefficient Resource Utilization

Following are the ways to solve:

- Route optimization
- Path symmetry
- Tunnelling or capsulation
- Quality of services
- Traffic engineering

4) What are the differences between the CoA handling in the FA vs. handling in the MN itself?

The care-of-address (CoA) handling in Mobile IP can differ depending on whether it is managed by the Foreign Agent (FA) or the Mobile Node (MN) itself. Here are the main differences between the two approaches.

the main differences in CoA handling between the Foreign Agent and the Mobile Node itself lie in how the CoA is assigned, managed, and updated. With a Foreign Agent, the CoA is assigned by the Foreign Agent and encapsulated packets are routed through the Foreign Agent. Without a Foreign Agent, the mobile node manages its own CoA, encapsulates outgoing packets with its CoA, and sends them directly to the Home Agent for routing

5) Which optimizations can be made in order to smooth data transport after a handover?

- Pre-Handover Preparation
- Fast Handover Techniques
- Predictive Handover
- Buffering and Reordering
- Quality of Service (QoS) Prioritization
- Dynamic Resource Allocation
- Parallel Data Paths
- Localized Handover Management

6) What is the purpose of reverse tunnelling? In which case is it helpful?

Reverse tunnelling is a technique used in networking to establish a connection from a destination (such as a server) back to the source (such as a client), even when the destination is behind a firewall or NAT (Network Address Translation) device.

The purpose of reverse tunneling is to allow remote access to devices or services that are not directly accessible from the internet due to security measures like firewalls or NAT configurations. By initiating the connection from the destination to the source, reverse tunneling enables bi-directional communication between the two devices, effectively bypassing any restrictions imposed by the network.

It is helpful in:

Remote administration

Secure Access to internal devices

IOT devices

7) What is the purpose of DHCP? What problem does it solve?

The Dynamic Host Configuration Protocol (DHCP) is a network protocol used to automatically assign IP addresses and other network configuration parameters to devices on a network.

Its primary purpose is to simplify the process of network configuration and management, particularly in large networks, by automating the assignment of IP addresses and reducing the need for manual configuration.

It solves following problems:

IP address management

Address conflicts

Configuration Consistency

Scalability

Network Layer MM – Advanced solutions:

1) Identify the limitations of Mobile IP. Which optimizations can you think of?

Limitations of Mobile IP:

- Tunnelling overhead
- Handover Latency
- Dependency on Home Agent
- Sub optimal Routing

Here are the optimizations which can be done:

- Localized Routing
- Predictive handover
- Distributed Mobility Management
- IP Address resolution Caching

2) What is the purpose of these optimizations?

Localized Routing

The purpose is to improve network efficiency and reduce latency by enabling direct communication between the mobile node and its corresponding nodes in the foreign network, bypassing the home agent whenever feasible

Predictive handover

This optimization aims to minimize handover latency and maintain seamless connectivity by proactively initiating handover procedures based on predictions of the mobile node's movement, reducing disruptions in communication.

Distributed Mobility Management

The purpose is to enhance scalability, fault tolerance, and load distribution in the network by distributing mobility management functions across multiple network nodes rather than relying solely on a centralized home agent.

IP Address resolution Caching

This optimization aims to minimize signalling overhead and handover latency by caching IP address mappings, reducing the frequency of location updates between the mobile node and the home agent, and improving overall efficiency.

3) Which problem is addressed by Hierarchical MIP?

Hierarchical Mobile IP (HMIP) addresses the scalability issue in traditional Mobile IP deployments, especially in large-scale networks with numerous mobile nodes. HMIP solves this problem by introducing a hierarchical architecture where mobility management functions are distributed across multiple levels of hierarchy.

4) What is the idea of HMIP?

The idea behind Hierarchical Mobile IP (HMIP) is to improve the scalability and efficiency of Mobile IP deployments, especially in large-scale networks with a significant number of mobile nodes. HMIP achieves this by organizing mobile nodes into hierarchical groups and distributing mobility management functions across multiple levels of hierarchy.

In HMIP, the network is divided into multiple domains or regions, with each domain containing a hierarchical structure of mobility agents. These mobility agents include local mobility agents (LMAs) and regional mobility agents (RMAs).

5) Which problem is addressed by Cellular IP? What is the idea of Cellular IP?

Cellular IP addresses the issue of mobility management and handover latency in wireless cellular networks, particularly in scenarios where mobile devices move between different cells or base stations.

The idea behind Cellular IP is to optimize the handover process and improve the efficiency of communication for mobile devices within a cellular network.

This is achieved by:

Cellular Structure

Localized Routing

Fast Handover

6) Why do updates of the routing table make sense here?

Updates of the routing table make sense in Cellular IP because they enable mobile devices to maintain accurate and up-to-date information about the network topology and available routes.

Updating the routing table allows mobile devices to dynamically adapt to changes in their surroundings and select the most efficient routes for communication.

7) Can Cellular IP be generalized to non-hierarchical systems?

Here's how Cellular IP can be generalized to non-hierarchical systems:

- Dynamic Routing
- Neighbour Discovery
- Fast Handover
- Adaptive Transmission Power

8) What is the idea of Proxy MIP? Which problem is addressed by Proxy MIP?

The idea of Proxy Mobile IP (PMIP) is to streamline mobility management in IP-based wireless networks by introducing a proxy entity known as the Local Mobility Anchor (LMA). The LMA serves as a proxy for the mobile device's home agent, managing its mobility within the access network.

This approach reduces handover latency and signaling overhead by offloading mobility management functions to the LMA, enhancing the efficiency of mobility management in wireless networks.

